

Charotar University of Science and Technology, CHANGA**Second Semester of M.Tech(EVD/CSE) Theory Exam****May 2018****Subject: EC764 Real Time Operating System****Date: 03/05/2018 (Thursday)****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70**

- Instructions:** (1) All questions are compulsory.
 (2) Indicate clearly, the option you attempt along with its respective question number.
 (3) Figures to the right indicate **full** marks.
 (4) Make suitable assumptions and draw neat figures wherever if required.
 (5) Section I and II must be attempted in separate answer sheets.

Section - I

- Q.1 Answer the questions below. [07]**
- a.** When several processes access the same data concurrently and the outcome of the execution depends on the particular order in which the access takes place, is called **[01]**
 i) dynamic condition ii) race condition
 iii) essential condition iv) critical condition
- b.** Which one of the following is a synchronization tool? **[01]**
 i) thread ii) semaphore
 iii) pipe iv) socket
- c.** The jacketing technique is used to **[01]**
 i) convert a blocking system call into nonblocking system call
 ii) create a new thread
 iii) communicate between threads
 iv) terminate a thread
- d.** The _____ maintains pending and blocked bit vectors in context of each process. **[01]**
 i) CPU ii) Memory
 iii) Process iv) Kernel
- e.** In UNIX, the abort() function sends the _____ signal to the calling process, causing abnormal termination. **[01]**
 i) SIGTERM ii) SIGSTOP
 iii) SIGABORT iv) SIGABRT
- f.** Using EDF algorithm practically, it is impossible to achieve 100 percent utilization due to: **[01]**
 i) interrupt handling
 ii) the cost of context switching
 iii) power consumption
 iv) all of the mentioned
- g.** The _____ scheduling algorithm schedules periodic tasks using a static priority policy with preemption. **[01]**
 i) earliest deadline first ii) rate monotonic
 iii) first cum first served iv) priority

Q.2(a) Three processes with process IDs p1,p2 and p3 with estimated execution time 8,4,7 milliseconds respectively enters the ready queue together in order p1,p3,p2.Process p4 with estimated execution time 5 milliseconds entered in the ready queue 5 milliseconds later the start of execution of p1.Calculate the waiting time and Turnaround time for each process and Average waiting time and Average Turnaround time in RR algorithm with time slice=3ms. **[07]**

OR

Q.2(a) Three processes with process IDs p1,p2 and p3 with estimated execution time 12,14,8 milliseconds respectively enters the ready queue together .Process p4 with estimated execution time 4 milliseconds entered in the ready queue after 3 milliseconds. Calculate the waiting time and Turnaround time for each process and Average waiting time and Average Turnaround time in SRT Preemptive algorithm. **[07]**

Q.2(b) Three processes with process IDs p1,p2, p3 and p4 with estimated execution time 7,8,12,6 milliseconds and priorities 1,0,4,3 (0-highest ,4-lowest) respectively enters the ready queue together .Process p5 with estimated execution time 5 milliseconds and priority 2 entered in the ready queue after 5 milliseconds. Calculate the waiting time and Turnaround time for each process and Average waiting time and Average Turnaround time in Priority based Preemptive algorithm. **[07]**

Q.3 Attempt any two. **[14]**

(a) What is priority inversion? What are the different technique adopted for handing priority inversion? **[07]**

(b) List Real Time Operating System(RTOS) available in market. Explain the different functional and non-functional requirements that needs to be evaluated in the selection of an RTOS. **[07]**

(c) What is role of Inter Process communication to develop RTOS? List IPC mechanism supported by RTOS. With help of suitable example demonstrate role of it to develop real time system. **[07]**

Section - II

Q.4 Do as directed. **[07]**

a. VxWorks is centered around **[01]**

- | | |
|---------------------|---------------------------|
| i) wind microkernel | ii) linux kernel |
| iii) unix kernel | iv) none of the mentioned |

b. The first process launched by the linux kernel is **[01]**

- | | |
|--------------------|--------------------|
| i) init process | ii) zombie process |
| iii) batch process | iv) boot process |

c. Which one of the following is not shared by threads? **[01]**

- | | |
|-------------------------------------|---------------------------|
| i) program counter | ii) stack |
| iii) both program counter and stack | iv) none of the mentioned |

d. There are two processes P1 and P2, whose periods are 50 and 100 respectively. P1 is assigned higher priority than P2. The processing times are $t_1 = 20$ for P1 and $t_2 = 35$ for P2. Is it possible to schedule these tasks so that each meets its deadline using Rate monotonic scheduling ? **[01]**

- | | |
|------------|---------------------------|
| i) yes | ii) no |
| iii) maybe | iv) none of the mentioned |

e. In UNIX, the _____ system call is used to send a signal. **[01]**

- | | |
|-----------|-------------|
| i) sig | ii) send |
| iii) kill | iv) sigsend |

- f. If the resources are always preempted from the same process, _____ can occur. [01]
 i) deadlock ii) system crash
 iii) aging iv) starvation
- g. To avoid deadlock [01]
 i) there must be a fixed number of resources to allocate
 ii) resource allocation must be done only once
 iii) all deadlocked processes must be aborted
 iv) inversion technique can be used
- Q.5(a) Develop a FreeRTOS code to retrieve and change the priority assigned to the task as per the following requirements [07]
 i) Create two tasks with names “Task1” and “Task2” respectively. The stack size for both the task are 240
 ii) Priority for Task1 is higher than Task2.
 iii) Task1 is spawned first and then Task2.
 iv) Task1 retrieves its priority and print it on console and then raise the priority of Task2.
 Task2 retrieves its priority and print it on console and then lower its priority.
- OR**
- Q.5(a) Develop a FreeRTOS code to retrieve and change the priority assigned to the task as per the following requirements [07]
 i) Create Binary Semaphore.
 ii) Create Task1 as HandlerTask with the priority 3 and stack size 240.
 iii) Create Task2 as PerodicTask with the priority 1 and stack size 240.
 iv) Execute ProdicTask First. Periodic Task raise the Software Interrupt.
 v) Software Interrupt executes, release binary semaphore and context switch is transferred to higher priority task as HandlerTask.
 HandlerTask take semaphore and print it status on console.
- Q.5(b) Assume that freeRTOS utility is dumped in the Arduino Board. Develop an Arduino code for the following specification. [07]
 i) Create two tasks Task 1 & Task 2 as VsenderTask with the priority 1.
 ii) Create Task 3 as Receiver Task with Priority 2.
 iii) Create a Queue to hold maximum of 5 data of type long integer.
 iv) Send the data to the Queue using xQueueSendToBack API with no blocking time.
 v) Receive the data from the Queue using xQueueReceive API with 100 ms blocking time.
- Q.6 Attempt any two. [14]
- a. Which RTOS is use to design aero space type applications? How it differ from other RTOS? Draw and explain process life cycle of it. [07]
- b. With help of suitable example demonstrate task management of uCOS-II. [07]
- c. Develop a linux program for the following environment: Create two thread in which 1st thread write A....Z in “data” file whereas 2nd thread write a.....z in “data” file. Use the Mutex IPC to lock and unlock both the threads whenever it writes into “data” file. [07]
